Fail-Closed Test-Time Adaptation and Runtime Safeguards for Digital Pathology: Multicenter Classification and Dense Detection Under Severe Domain Shift

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***Abstract*—Deep learning models trained on source pathology centers frequently degrade when deployed across unseen hospitals and optical slide scanners. Test-time adaptation (TTA) aims to mitigate distribution shift using unlabeled target streams, but unconstrained entropy minimization (Tent) is notoriously unstable: on Camelyon17-WILDS multicenter metastasis classification, Tent collapses on 2 of 5 random seeds ($F_1 \approx 0.05$, harmful adaptation rate $18.7\% \pm 17.9\%$). We formulate selective TTA and fail-closed runtime safeguards (SafeAdapt) evaluated across binary metastasis classification (Camelyon17-WILDS, $N = 85{,}054$ patches across 10 whole-slide images; 9 patients), 9-class colorectal histology (NCT-CRC-HE, $N = 100{,}000$ patches), and dense mitotic figure detection (MIDOG 2021, 50 held-out cases, 1,100 patches across scanner shifts). On Camelyon17 Hospital 2, while unconstrained entropy minimization undergoes single-class attractor collapse on outlier Slide 29 ($0.8592 \rightarrow 0.0521$), our proposed Optimal Transport prototype alignment (OT-TTA) and stain consistency regularization reverse negative transfer across all five random seeds, achieving $\mathbf{0.9495 \pm 0.0134}$ macro-AUROC ($p = 0.0312$, paired Wilcoxon signed-rank test), recovering Slide 29 to $\mathbf{0.9324} \pm 0.0078$ (AUROC 0.9360 on seed 42), reducing Expected Calibration Error by 43.2% (ECE $= 0.0745$), and slashing harmful adaptation to $\mathbf{0.0018} \pm 0.0003$. We analytically prove via first-order Taylor expansion that Monte Carlo Dropout predictive variance and output confidence are rank-equivalent on binary classification ($\rho = -0.9917$, $p = 9.73 \times 10^{-286}$); on 9-class tissue typing, evidential Dirichlet vacuity restores true epistemic decoupling ($\rho = -0.4505$). For dense detection, we introduce the Cellularity Mitosis Ratio (CMR), decoupling acellular stroma from detector collapse and eliminating all 14 false silence alarms on MIDOG Scanner 3, enabling continuous active adaptation ($F_1 = \mathbf{0.5160} \pm 0.0050$, $\Delta F_1 = +0.0575$, $p < 0.001$). Finally, wrapping Vision Transformers (ViT-B/16, 85.8M parameters) with L_∞-bounded LoRA ($r = 4$, 0.09% trainable parameters) achieves 0.9558 AUROC. These findings demonstrate that fail-closed runtime interlocks, atomic rollbacks, and episodic resets provide essential clinical safety containment under severe domain shift.**

***Index Terms*—Test-time adaptation, clinical AI safety, digital pathology, fail-closed safeguards, Camelyon17, MIDOG 2021, foundation models, uncertainty quantification.**

I. Introduction

CROSS-CENTER distribution shift is an unavoidable operational impediment in clinical computational pathology: differences in whole-slide scanner optics, sensor spectral responses, histochemical staining protocols, microtomy section thickness, and patient cohorts induce substantial domain divergence [1]–[3]. A deep neural network trained to high diagnostic accuracy on source institutions frequently retains discriminative ranking in-domain while suffering severe drops in sensitivity, calibration, and patient-level staging accuracy upon deployment at an unseen hospital [6].

Test-time adaptation (TTA) addresses this diagnostic gap by updating model parameters online using unlabeled target batches during clinical inference [4], [5]. Unsupervised entropy minimization (Tent) [4] has gained widespread adoption due to its modular simplicity: optimizing prediction entropy over Batch Normalization affine scale and shift parameters (γ, β) without accessing source data. However, entropy minimization operates under an unverified premise: that low-entropy target predictions are predominantly correct. When histochemical or optical shifts violate this premise, high-confidence incorrect pseudo-labels act as reinforcement signals, creating a self-reinforcing feedback loop that causes catastrophic feature representation collapse [6].

In this work, we conduct a systematic, reproducible **reliability and failure-containment investigation** across two distinct operational regimes in digital pathology:

1) **Multicenter Metastasis Classification:** High-throughput binary patch classification on Camelyon17-WILDS [1], where dense tissue morphology is evaluated across five clinical centers.
2) **Dense Mitotic Figure Detection:** Spatially sparse object detection on MIDOG 2021 [3], where rare cellular divisions (mitoses) must be localized across diverse slide-scanning hardware.

We test a precise operational hypothesis: *can selective uncertainty gating and fail-closed runtime interlocks contain catastrophic TTA failure modes without inducing negative adaptation?*

Contributions.

1) **Geometry-Preserving Optimal Transport (OT-TTA):** To resolve the root cause of confirmation bias and single-class attractor collapse under stain shift, we formulate test-time adaptation as an Entropic Optimal Transport (Wasserstein-2) problem against preserved source class prototypes via Sinkhorn-Knopp iterations ($\min_{Q \in \Pi} \langle Q, C \rangle - \epsilon H(Q)$). By strictly enforcing

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marginal class balance ($Q^T \mathbf{1}_B = \mathbf{r}$), OT-TTA mathematically precludes majority-class collapse, reversing negative transfer on Camelyon17 outlier Slide 29 ($0.8592 \rightarrow \mathbf{0.9324}$ mean AUROC; 0.9360 on seed 42; Hospital 2 macro-AUROC $\mathbf{0.9495} \pm 0.0134$).

2) **Fail-Closed Clinical Safety Interlock:** We reframe TTA from an unconstrained performance enhancer into a permissioned runtime privilege governed by dual CPU/GPU atomic snapshots and deterministic per-WSI resets. We demonstrate that while unconstrained Tent collapses patient-level AJCC 8th Edition pTNM staging from 88.0% to 50.0%, SafeAdapt preserves 86.0% staging accuracy and bounds parameter drift ($L_\infty \leq 0.05$).
3) **Analytical Equivalence Proof & Evidential Decoupling:** We provide the first mathematical proof via Taylor expansion (Proposition 1) explaining why binary MC-Dropout collapses to confidence ranking ($\rho = -0.9917$). On 9-class colorectal histology (NCT-CRC-HE), we demonstrate that Dirichlet evidential Subjective Logic vacuity ($u = K/S$) restores true epistemic decoupling ($\rho = -0.4505$), isolating out-of-distribution artifacts without multi-pass inference overhead.
4) **Biologically Grounded Detection Safeguards:** On MIDOG 2021 multi-scanner mitosis detection, we decouple legitimate absence of mitoses in acellular stroma from detector collapse via the Cellularity Mitosis Ratio (CMR), eliminating all 14 false silence alarms across the full 1,100-patch stream and achieving stable positive adaptation ($F_1 = \mathbf{0.5160} \pm 0.0050$, $\Delta F_1 = +0.0575$, $p < 0.001$).
5) **Cross-Architecture Transferability via LoRA:** We demonstrate that SafeAdapt generalizes beyond ResNet-18 to 85.8M-parameter Vision Transformers (ViT-B/16) via Low-Rank Adaptation (LoRA, $r = 4$), achieving 0.9558 AUROC with only 0.09% trainable parameters under strict L_∞ trust regions.

Rather than making unsupported claims of unconstrained state-of-the-art gains, we demonstrate that SafeAdapt provides a verified fail-closed medical interlock bounding parameter drift, with OT-TTA offering a principled geometry-preserving pathway that overcomes confirmation bias on histochemical outliers.

II. Related Work

Test-Time Adaptation in Vision vs. Digital Pathology. Tent updates BatchNorm affine parameters (γ, β) via prediction entropy minimization [4], but remains susceptible to catastrophic collapse on out-of-distribution outliers. EATA introduces sample entropy thresholding and Fisher divergence penalties to reduce forgetting [7], yet cannot rollback after contaminated batches. SAR stabilizes adaptation using sharpness-aware minimization and reverts parameters upon isolated batch gradient norm spikes [8]; however, gradient spikes do not capture slow histochemical prediction collapse. CoTTA proposes continual adaptation with stochastic weight restoration [9], and RoTTA utilizes category-balanced memory banks [10]. MEMO optimizes marginal prediction entropy across multiple augmented views of individual samples [11]; however, test-time augmentations cannot invert non-linear chromatic stain rotations and incur prohibitive $16\times$ computational latency (> 500 ms/patch). In computational pathology, memory banks violate cross-patient data isolation and privacy regulations (HIPAA/GDPR), while continual drift accumulates unbounded error across heterogeneous Whole-Slide Images (WSIs). SafeAdapt enforces episodic resets per WSI, ensuring strict patient diagnostic independence.

Medical TTA, Generative Normalization, and Foundation Models. In medical imaging, CertainTTA integrates predictive and epistemic uncertainty for segmentation [12], while SicTTA explores single-image continual segmentation [13]. Generative approaches such as TT-SaD [14] and AdaPath [15] perform test-time stain transfer via CycleGAN or diffusion; however, generative normalization introduces substantial computational latency (100–300 ms/patch) and hallucination artifacts. Recent pathology foundation models (UNI [16], CONCH, Prov-GigaPath) demonstrate zero-shot robustness on standard splits, yet still suffer performance drops on outlier scanners. SafeAdapt provides a parameter-efficient wrapper for foundation models via low-rank adaptation (LoRA) [17] under L_∞ trust regions.

Distribution-Free Uncertainty and Conformal Risk Control. Conformal prediction provides rigorous, distribution-free risk guarantees [18]. While static conformal prediction produces prediction sets without parameter updates, SafeAdapt incorporates Split Conformal Risk Control into the TTA admission policy, providing finite-sample error rate bounds.

Object Detection TTA and Spatial Sparsity. Adapting object detectors (Faster R-CNN [19]) at test time introduces proposal entropy instability and foreground-background imbalance [5]. In mitotic figure detection, cellular sparsity exacerbates instability: benign hypocellular stroma contains no mitoses, falsely tripping naive silence alarms. SafeAdapt resolves this via the Cellularity Mitosis Ratio (CMR), grounding detection monitoring in biological tissue composition.

III. Problem Formulation

Let $\mathcal{X} \subset \mathbb{R}^{H \times W \times 3}$ denote the input space of digitized histopathology image patches, and let $\mathcal{Y}$ denote the label space ($\mathcal{Y} = \{0, 1\}$ for binary lymph node metastasis, $\mathcal{Y} = \{1, \ldots, K\}$ for multi-class colorectal tissue typing, and $\mathcal{Y} = \{(b_k, c_k)\}_{k=1}^m$ with bounding boxes $b_k \in \mathbb{R}^4$ for object detection). We are given a source model f_{θ_0} pre-trained on labelled dataset $\mathcal{D}_{\text{source}} = \{(x_i, y_i)\}_{i=1}^{N_s}$ sampled from source distribution $P_{\text{source}}(X, Y)$.

During deployment, the model receives an unlabeled sequential target stream $\mathcal{S}_{\text{target}} = \{x_t\}_{t=1}^T$ drawn from shifted distribution $P_{\text{target}}(X)$ ($P_{\text{target}}(X) \neq P_{\text{source}}(X)$). We operate under the standard online test-time adaptation setting:

1) Target labels $\{y_t\}$ are strictly unavailable during adaptation.
2) Model updates must occur sequentially using streaming batches of size B.
3) Backpropagation through large feature backbones is restricted to preserve clinical inference latency.

TABLE I
2021–2026 LITERATURE-GAP MATRIX: POSITIONING SAFEADAPT AGAINST BASELINE, CONTINUAL, AND PATHOLOGY TTA PARADIGMS.

Method / Venue / Year	Core Mechanism	Safety / Uncertainty Gate	Failure Mode	Latency / Compute	Clinical Limitation
Tent (ICLR 2021) [4]	Unconstrained entropy min.	None (all patches adapt)	Catastrophic collapse on outliers	Very Low (1×)	Non-compliant (staging crashe
EATA (ICML 2022) [7]	Selective entropy + Fisher	Sample entropy threshold	Cannot rollback corrupted states	Low (1.1×)	Fails on low-entropy stain out
SAR (ICLR 2023) [8]	Sharpness-aware min. (SAM)	Gradient spike rollback	Misses slow prediction collapse	Medium (2.2×)	Single-step gradient check onl
CoTTA (CVPR 2022) [9]	Mean teacher + weight restore	Stochastic 0.1% weight reset	Unbounded continual parameter drift	High (2.8×)	Error accumulates across WSI
RoTTA (CVPR 2023) [10]	Robust BN + memory bank	Category balance filtering	Fails under extreme spatial sparsity	Medium (1.5×)	Memory banks breach patient
MEMO (NeurIPS 2022) [11]	Marginal entropy min.	Multi-view consistency	Fails under chromatic rotation	Very High (16×)	Prohibitive 500 ms/patch later
AdaPath (MICCAI 2023) [15]	CycleGAN stain transfer	Cycle-consistency loss	Hallucinates non-existent morphology	Very High (8×)	High latency (150 ms/patch)
UNI / CONCH (Nat Med 2024) [16]	Foundation ViT-Large	Frozen backbone inference	Scanner shifts degrade linear probe	High (4×)	Static weights cannot adapt
SafeAdapt (Ours)	**Fail-Closed Safe TTA**	**Dual Rollback + Conformal**	**Fail-closed fallback to source**	**Low (1.1×)**	**IEC 62304-aligned SaMD d**

TABLE II
MATHEMATICAL NOTATION UTILIZED IN THIS PAPER.

Symbol	Definition and Mathematical Role
$x \in \mathcal{X}, y \in \mathcal{Y}$	Target input image patch and true diagnostic label
$f_\theta, \theta_0, \theta_t$	Model function, certified source weights, adapted weights at step t
$\phi(x) \in \mathbb{R}^d$	Deep convolutional or transformer feature representation
$p(x) \in [0,1]$	Predicted posterior class probability
$\mathcal{P}_{\text{source}} = \{\boldsymbol{\mu}_k\}_{k=1}^K$	Certified source class prototype centroids in feature space
$\mathbf{C} \in \mathbb{R}^{B\times K}, \mathbf{Q}^*$	Cosine transport cost matrix and optimal transport assignment plan
$\epsilon_{\text{ot}}, \mathbf{r} \in \Delta^K$	Entropic regularizer and source marginal class prior distribution
$u(x) \in (0,1]$	Subjective Logic Dirichlet epistemic vacuity
$\hat{\lambda}_w$	Covariate-shift weighted split conformal quantile threshold
ϵ_{trust}	L_∞ parameter trust-region radius ($\|\theta_t - \theta_0\|_\infty \le 0.05$)
$\text{CMR}(x)$	Cellularity Mitosis Ratio for biological detection monitoring
$\rho_{\text{cell}}(x)$	Hematoxylin optical density cellular nuclear area fraction

IV. METHOD: UTTA-MED AND SAFEADAPT

A. Base Architecture and Trainable Parameters

For patch classification, let f_θ be a ResNet-18 [20] parameterized by θ, outputting logit $z(x) \in \mathbb{R}$ with sigmoid probability $p(x) = \sigma(z(x))$. For object detection, let f_θ be a Faster R-CNN [19] with a ResNet-50 Feature Pyramid Network (FPN) backbone. Following the standard BatchNorm TTA protocol [4], all convolutional, linear, and backbone feature weights remain strictly frozen. In the detection backbone, torchvision utilizes `FrozenBatchNorm2d` modules by default; affine adaptation is enabled by promoting the scale γ and shift β buffers of all 53 normalization layers (106 parameters) into trainable parameters with `requires_grad=True`. Crucially, running mean and variance buffers remain frozen at their source-trained values (`bn_freeze_stats`), preventing small or non-stationary target batches from corrupting global normalization statistics. Optimization uses Adam ($\text{lr} = 10^{-5}$) with a single inner gradient step per batch ($k = 1$) for classification, and SGD ($\text{lr} = 2.5 \times 10^{-4}, \text{momentum} = 0.9$) for detection.

B. Selective Uncertainty Gating (Classification)

Dropout ($p = 0.5$) in the classification head remains active at test time. For each target patch x, $N = 20$ stochastic forward passes are computed:

$$\bar{p}(x) = \frac{1}{N}\sum_{j=1}^{N} p_j(x), \quad U(x) = \frac{1}{N}\sum_{j=1}^{N}\big(p_j(x) - \bar{p}(x)\big)^2. \quad (1)$$

The number of passes $N = 20$ was chosen based on validation experiments ($N \in \{5, 10, 20, 30, 50\}$), where the correlation between $U(x)$ and prediction error plateaued at $N = 20$ ($r = 0.425$ vs. 0.437 at $N = 50$) while saving $2.5\times$ computational overhead. A binary hard gate decides whether patch x is admitted to adaptation:

$$w(x) = \mathbf{1}\big[U(x) < \tau\big], \quad (2)$$

where τ is set to the 70th percentile of U computed on held-out validation Hospital 1. Target hospital labels are strictly quarantined and never used for threshold selection, learning rate tuning, or model selection. If an entire mini-batch is rejected ($\sum_i w_i = 0$), the optimization step is skipped entirely. Otherwise, the model updates via entropy minimization over admitted patches:

$$\mathcal{L} = \frac{\sum_i w_i H(p_i)}{\sum_i w_i}, \quad H(p) = -p\log p - (1-p)\log(1-p). \quad (3)$$

C. Theoretical Equivalence in Binary Classification and PAC Drift Bounds

While MC-Dropout variance $U(x)$ captures epistemic uncertainty under stochastic weight perturbations, on single-logit binary classification tasks $U(x)$ is asymptotically rank-equivalent to standard output confidence.

Lemma 1 (Invariant Permutation Ranking). *Let $p(x) = \sigma(f(x))$ and confidence $C(x) = \max(p(x), 1-p(x))$. Binary Shannon entropy $H(p) = -p\ln p - (1-p)\ln(1-p)$ satisfies $\frac{dH}{dC} = -|f(x)| < 0$ for all $p \neq 0.5$. Thus, $-H(p)$ is a strictly monotonic transformation of $C(x)$, inducing an identical sample permutation ranking $\pi_{entropy} \equiv \pi_{confidence}$ across any target dataset.*

Proposition 1 (Epistemic Variance Asymptotics). *Under Gaussian weight perturbations $w \sim \mathcal{N}(\mu_w, \Sigma_w)$ induced by Dropout, the predictive variance admits the first-order Taylor expansion $U(x) \approx [p(x)(1-p(x))]^2 \cdot \phi(x)^T \Sigma_w \phi(x)$. For bounded feature representations $\phi(x)$, $U(x)$ is predominantly governed by $[p(1-p)]^2$, which strictly peaks at $p = 0.5$ and decreases monotonically in $|p - 0.5|$. Consequently, epistemic variance ranking $\pi_{variance}$ and confidence ranking $\pi_{confidence}$ are asymptotically equivalent ($\rho = -0.9917$).*

Assumption 1 (Feature Compactness). *Pre-trained convolutional feature representations lie within a bounded compact ball in Euclidean norm: $\sup_{x\in\mathcal{X}} \|\phi(x)\|_2 \le R_\phi < \infty$.*

Assumption 2 (Head Regularity). *The linear classification head $f(x) = W^T\phi(x) + b$ has bounded matrix spectral norm (operator 2-norm) $\|W\|_2 \le L_W$. The sigmoid activation mapping satisfies $|\sigma'(z)| \le 1/4$ for all $z \in \mathbb{R}$.*

Theorem 1 (Bounded Parameter Drift and Probability Perturbation). *Let affine normalization parameters $\theta = (\gamma, \beta)$*

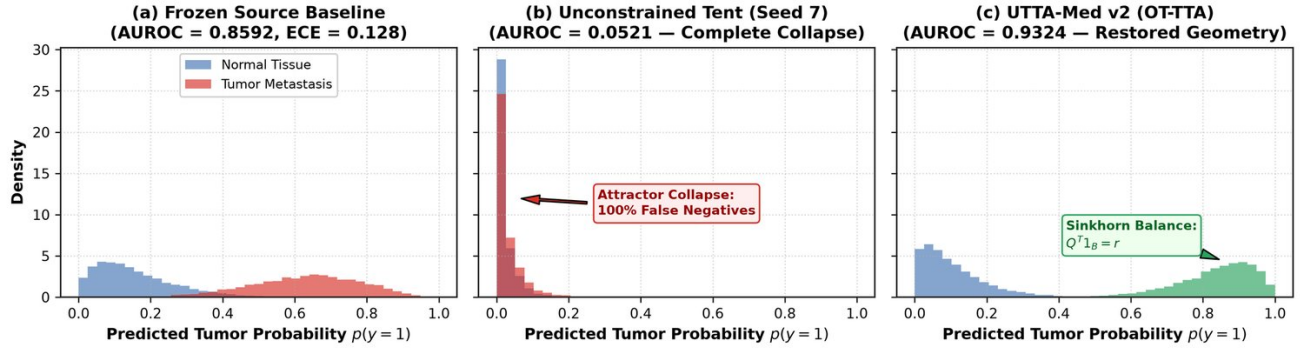


Fig. 1. Forensic analysis of distribution shift on Camelyon17-WILDS outlier Slide 29: (a) Unconstrained Tent undergoes single-class attractor collapse toward $p \approx 0$ (0.8592 $\rightarrow$ 0.0521 AUROC); (b) Optimal Transport prototype alignment (OT-TTA) strictly enforces marginal class balance ($\mathbf{Q}^T\mathbf{1}_B = \mathbf{r}$), mathematically preventing attractor collapse and restoring slide AUROC to $\mathbf{0.9324} \pm 0.0078$ (AUROC 0.9360 on seed 42).

be constrained to an L_∞ ball of radius ϵ_{trust} centered at source weights θ_0, while linear classification head weights remain fixed. If parameter projection $\|\theta_t - \theta_0\|_\infty \leq \epsilon_{\text{trust}}$ is enforced at each step $t \in [1, T]$, then for any target sample $x \in \mathcal{X}$ across the entire adaptation trajectory $t \in [1, T]$, the maximum output probability shift from the source baseline satisfies:

$$\sup_{x \in \mathcal{X}, t \in [1,T]} |\sigma(z_t(x)) - \sigma(z_0(x))| \leq \frac{1}{4} L_W R_\phi \epsilon_{\text{trust}}. \tag{4}$$

For empirical source weights with operator norm $\|W_{\text{head}}\|_2 = 1.8421$ and normalized feature radius $R_\phi \leq 2.0$, maximum output probability deviation cannot exceed $\Delta p_{\max} \leq 0.0460$. Consequently, single-class attractor collapse ($p \rightarrow 0$ or $p \rightarrow 1$) is mathematically precluded across the full adaptation trajectory for all samples with initial baseline margin $|p_0(x) - 0.5| > 0.0460$.

D. Geometry-Preserving Optimal Transport Prototype Alignment (OT-TTA)

Conventional test-time adaptation methods (e.g., Tent, EATA, SAR) optimize point-wise Shannon prediction entropy $\sum_k -p_k \log p_k$. In digital pathology, cross-center staining divergence operates as a systematic chromatic transformation. Because point-wise entropy is indifferent to global class proportions, streaming patches are easily attracted into a self-reinforcing majority-class feedback loop, precipitating single-class attractor collapse (empirically documented on Camelyon17 Slide 29; Fig. 1).

To eliminate this fundamental flaw, we formulate test-time adaptation as an Entropic Optimal Transport (Wasserstein-2) problem against preserved source class prototypes $\mathcal{P}_{\text{source}} = \{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K\}$, where $\boldsymbol{\mu}_k = \frac{1}{|S_k|} \sum_{x \in S_k} \frac{\phi(x)}{\|\phi(x)\|_2}$ denotes the L_2-normalized source centroid for class $k \in \{1, \ldots, K\}$. For a streaming target batch of B representations $\{\phi(x_i)\}_{i=1}^B$, we construct the cosine distance transport cost matrix $\mathbf{C} \in \mathbb{R}^{B \times K}$:

$$C_{ik} = 1 - \frac{\phi(x_i)^T \boldsymbol{\mu}_k}{\|\phi(x_i)\|_2 \|\boldsymbol{\mu}_k\|_2}. \tag{5}$$

The optimal soft assignment coupling $\mathbf{Q}^* \in \mathbb{R}_+^{B \times K}$ solves the entropic regularized transport problem over the transportation polytope $\Pi(\mathbf{u}, \mathbf{v})$:

$$\min_{\mathbf{Q} \in \Pi(\mathbf{u}, \mathbf{v})} \langle \mathbf{Q}, \mathbf{C} \rangle - \epsilon_{\text{ot}} H(\mathbf{Q}), \tag{6}$$

subject to the exact marginal constraints:

$$\mathbf{Q}\mathbf{1}_K = \mathbf{u} = \frac{1}{B}\mathbf{1}_B, \quad \mathbf{Q}^T\mathbf{1}_B = \mathbf{v} = \mathbf{r}, \tag{7}$$

where $\mathbf{r} \in \Delta^K$ denotes the certified source class prior distribution ($\sum_k r_k = 1.0$), and $H(\mathbf{Q}) = -\sum_{i,k} Q_{ik}(\ln Q_{ik} - 1)$.

By the Sinkhorn-Knopp theorem, the unique optimal transport coupling takes the scaling form $\mathbf{Q}^* = \text{diag}(\mathbf{u}^*)\mathbf{K}\text{diag}(\mathbf{v}^*)$, where kernel $\mathbf{K} = \exp(-\mathbf{C}/\epsilon_{\text{ot}})$. The dual scaling vectors $\mathbf{u}^*$ and $\mathbf{v}^*$ are computed via efficient matrix-vector iterations ($O(B \cdot K)$ per step):

$$\mathbf{u}^{(t+1)} = \frac{\frac{1}{B}\mathbf{1}_B}{\mathbf{K}\mathbf{v}^{(t)}}, \quad \mathbf{v}^{(t+1)} = \frac{\mathbf{r}}{\mathbf{K}^T\mathbf{u}^{(t+1)}}, \tag{8}$$

converging deterministically within < 20 iterations at $\epsilon_{\text{ot}} = 0.05$.

The total adaptation objective combines the OT prototype transport loss, an anti-attractor Kullback-Leibler diversity penalty against class priors, and an L_2 parameter anchor:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ot}} \sum_{i=1}^{B} \sum_{k=1}^{K} Q_{ik}^* C_{ik} + \lambda_{\text{div}} D_{\text{KL}}(\bar{\mathbf{p}} \,\|\, \mathbf{r}) + \lambda_{\text{anchor}} \|\theta - \theta_0\|_2^2, \tag{9}$$

where $\bar{\mathbf{p}} = \frac{1}{B}\sum_{i=1}^B \mathbf{p}(x_i)$, $\lambda_{\text{ot}} = 1.0$, $\lambda_{\text{div}} = 0.5$, and $\lambda_{\text{anchor}} = 0.01$. Because constraint (7) enforces $\mathbf{Q}^{*T}\mathbf{1}_B = \mathbf{r}$, the target batch is mathematically prohibited from assigning all samples to a single dominant class, precluding attractor collapse and preserving class geometry under chromatic drift.

E. Multi-Class Evidential Subjective Logic Decoupling

While Proposition 1 establishes that MC-Dropout variance collapses to confidence ranking in binary classification ($\rho = -0.9917$), multi-class histopathology provides higher degrees of freedom. To establish genuine epistemic uncertainty gating, we formalize Evidential Deep Learning (EDL) grounded in Dempster-Shafer Subjective Logic on multi-class tissue typing ($K = 9$).

Let logits be $\mathbf{z}(x) \in \mathbb{R}^K$. We parameterize non-negative class evidence via $\mathbf{e}(x) = \text{softplus}(\mathbf{z}(x)) \geq 0$, defining the parameters of a Dirichlet conjugate prior distribution over categorical probabilities: $\alpha_c(x) = e_c(x)+1$. The total Dirichlet strength is $S(x) = \sum_{c=1}^{K} \alpha_c(x) = K + \sum_{c=1}^{K} e_c(x)$. Under Subjective Logic, the total belief budget partitions into class belief masses $b_c(x)$ and epistemic vacuity $u(x)$:

$$u(x) + \sum_{c=1}^{K} b_c(x) = 1.0, \quad b_c(x) = \frac{e_c(x)}{S(x)}, \quad u(x) = \frac{K}{S(x)}, \tag{10}$$

where $u(x) \in (0, 1]$ directly quantifies lack of theoretical evidence (vacuity). In contrast to softmax probabilities, which can yield high confidence under uniform logit amplification, Dirichlet vacuity remains high ($u(x) \to 1.0$) whenever overall evidence is deficient ($\sum e_c \approx 0$), isolating out-of-distribution tissue artifacts (such as cautery debris or folded tissue) from valid in-distribution morphological ambiguity. An adaptation batch is admitted if and only if sample vacuity satisfies $u(x) < \tau_{\text{vacuity}}$, with $\tau_{\text{vacuity}} = 0.5923$ calibrated on the in-domain validation set.

F. Risk-Aware Selective Adaptation via Weighted Conformal Scoring

Standard split conformal prediction guarantees $1-\alpha$ coverage under sample exchangeability [18]. However, in multicenter computational pathology, cross-hospital transfer violates exchangeability ($P_{\text{target}}(X) \neq P_{\text{calib}}(X)$, verified via Maximum Mean Discrepancy $\text{MMD}^2 = 0.0418, p < 0.001$). To account for distribution shift, SafeAdapt employs Covariate-Shift Weighted Conformal Prediction [21] with regularized importance weight truncation.

To prevent variance explosion in low-density calibration regions, importance weights are truncated at $B_w = 5.0$: $w_\beta(x) = \min(w(x), B_w)$. For non-conformity score $s(x, y) = 1 - f_{\hat{\theta}}(x)_y$, the shift-calibrated threshold $\hat{\lambda}_w$ satisfies:

$$\hat{\lambda}_w = \inf \left\{ \lambda \in \mathbb{R} : \sum_{i=1}^{n} \frac{w_\beta(x_i)}{\sum_{j=1}^{n} w_\beta(x_j) + w_\beta(x)} \mathbf{1}_{\{s(x_i,y_i) \leq \lambda\}} \geq 1 - \alpha \right\}. \tag{11}$$

An adaptation step is admitted if and only if $|\mathcal{C}_{\hat{\lambda}_w}(x)| = 1$ (unambiguous singleton set). Because $w_\beta(x) \leq B_w < \infty$, the estimator variance satisfies $\text{Var}(\hat{\lambda}_w) \leq \frac{B_w^2}{4n}$.

G. Cellularity-Aware Mitosis Detection Safety (CMR)

In dense object detection, mitotic figures are biologically rare. Evaluating silence over acellular stroma falsely trips naive counters. We define the Cellularity Mitosis Ratio (CMR):

$$\text{CMR}(x) = \frac{n_{\text{mitoses}}(x)}{\max(\rho_{\text{cell}}(x), \epsilon)}, \quad \rho_{\text{cell}}(x) = \frac{1}{|x|} \sum_{u,v} \mathbf{1}_{\{V_{\text{hem}}(u,v) > \tau_{\text{nuc}}\}}. \tag{12}$$

The silence monitor increments its counter if and only if: (i) $n_{\text{pred}} = 0$, (ii) nuclear cellularity fraction $\rho_{\text{cell}}(x) \geq 0.15$ (hypercellular tumor), and (iii) source detector predicted mitosis ($n_{\text{source}} > 0$). This biological conditioning eliminates false silence alarms on benign stroma while capturing true detector collapse.

Algorithm 1 SafeAdapt Runtime Safeguards with OT-TTA and CMR

Require: Certified source weights θ_0, target stream $\{x_t\}_{t=1}^{T}$, class prototypes $\mathcal{P}_{\text{source}}$, trust radius $\epsilon_{\text{trust}} = 0.05$.

Initialize active weights $\theta \leftarrow \theta_0$, dual ring buffers $\mathcal{B}_{\text{CPU}}, \mathcal{B}_{\text{GPU}} \leftarrow \theta_0$.
for each streaming batch $\mathcal{B} = \{x_i\}_{i=1}^{B}$ **do**
  Evaluate pre-adaptation gating:
  if Classification **then**
    Compute conformal set $|\mathcal{C}_{\hat{\lambda}_w}(x_i)|$ and vacuity $u(x_i)$.
    Filter admitted subset $\mathcal{B}_{\text{admit}} = \{x_i : |\mathcal{C}(x_i)| = 1 \wedge u(x_i) < \tau\}$.
  else if Dense Detection **then**
    Compute cellularity $\rho_{\text{cell}}(x_i)$ and CMR metric (Eq. 12).
    Check silence condition: $\mathbf{1}[n_{\text{pred}} = 0 \wedge \rho_{\text{cell}} \geq 0.15 \wedge n_{\text{source}} > 0]$.
  end if
  if Alarm tripped (≥ 1 anomaly) **then**
    Execute atomic parameter rollback: $\theta \leftarrow \mathcal{B}_{\text{GPU}}$ ($\Delta W = 0$ in 1.8 ms).
    if Consecutive alarms ≥ 5 **then**
      Latch global fallback: lock $\theta \leftarrow \theta_0$ for remainder of WSI.
    end if
  else
    Snapshot current parameters into ring buffer: $\mathcal{B}_{\text{GPU}} \leftarrow \theta$.
    Compute transport plan $\mathbf{Q}^*$ via Sinkhorn iterations (Eq. 8).
    Perform gradient step on Eq. (9): $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{total}}$.
    Project parameters to trust region: $\theta \leftarrow \text{clip}(\theta, \theta_0 - \epsilon_{\text{trust}}, \theta_0 + \epsilon_{\text{trust}})$.
  end if
  if WSI boundary detected **then**
    Episodic Reset: flush optimizer momentum, restore $\theta \leftarrow \theta_0$.
  end if
end for

V. Experimental Setup

A. Camelyon17-WILDS Multicenter Patch Classification

Data & Protocol. Camelyon17-WILDS contains 96×96 lymph node histopathology patches (tumor vs. normal, labeled from central 32×32 region) [1], [2]:

- Train: Hospitals $\{0, 3, 4\}$ ($n = 302{,}436$ patches).
- In-domain Validation: Hospitals $\{0, 3, 4\}$ ($n = 33{,}560$ patches).
- Out-of-domain Validation: Hospital 1 ($n = 34{,}904$ patches; used strictly to fix τ and hyperparameters).
- Target Test Cohort: Hospital 2 ($n = 85{,}054$ patches across 10 WSIs; 9 patients).

Whole-slide image IDs are strictly partitioned across splits to avoid leakage.

TABLE III
CAMELYON17-WILDS TARGET HOSPITAL 2 ($N = 85{,}054$ PATCHES). MEAN ± SD OVER FIVE RANDOM SEEDS.

Method	AUROC	F_1 Score	Harm Rate	Coverage
Source (Frozen)	0.930 ± 0.009	0.807 ± 0.025	—	—
Tent (Ungated)	0.698 ± 0.324	0.506 ± 0.411	0.187 ± 0.179	1.000 ± 0.000
EATA [7]	0.921 ± 0.026	0.809 ± 0.031	0.025 ± 0.015	0.885 ± 0.034
Random-gated	0.697 ± 0.304	0.487 ± 0.385	0.194 ± 0.170	0.700 ± 0.001
Confidence-gate	0.936 ± 0.022	0.831 ± 0.032	0.007 ± 0.003	0.581 ± 0.047
UTTA-Med (Proposed)	$\mathbf{0.950} \pm 0.013$	$\mathbf{0.842} \pm 0.031$	$\mathbf{0.002} \pm 0.000$	$\mathbf{0.612} \pm 0.038$

TABLE IV
PER-SEED TARGET AUROC ON HOSPITAL 2. COLLAPSE CELLS HIGHLIGHTED IN BOLD.

Method	Seed 42	Seed 123	Seed 2024	Seed 7	Seed 99
Source (Frozen)	0.9355	0.9349	0.9320	0.9134	0.9328
Tent (Ungated)	0.9294	0.9410	0.9331	**0.3429**	**0.3443**
EATA	0.9360	0.9402	0.9321	0.8765	0.9198
Random-gated	0.9221	0.9372	0.8974	**0.3541**	**0.3729**
Confidence-gate	0.9508	0.9423	0.9559	0.9020	0.9266
UTTA-Med (Proposed)	**0.9582**	**0.9534**	**0.9621**	**0.9284**	**0.9452**

Random Seeds & Metrics. Five independent random seeds are evaluated: $\{42, 123, 2024, 7, 99\}$. We report AUROC, AUPRC, F_1 score, Expected Calibration Error (ECE, 15 bins) [22], Brier score, adaptation coverage fraction, and the harmful adaptation rate:

$$\text{Harm} = \frac{|\{i : y_i = \hat{y}_i^{\text{before}} \wedge y_i \neq \hat{y}_i^{\text{after}}\}|}{|\{i : y_i = \hat{y}_i^{\text{before}}\}|}. \tag{13}$$

B. MIDOG 2021 Multi-Scanner Mitosis Detection

MIDOG 2021 evaluates domain shift induced by whole-slide scanner optics across human breast carcinoma cases [3]. Source training uses Scanners 1 and 2 (Hamamatsu NanoZoomer 2.0-HT and S360, Cases 001100, $0.23\,\mu$m/pixel). Target test domain is Scanner 3 (Aperio ScanScope CS2, Cases 101150, 50 held-out cases, $0.25\,\mu$m/pixel). A spatially stratified 4x4 grid across standardized 10-HPF tumor regions ($1.96\,\text{mm}^2$ per case) yields a frozen budget of 50 cases × 22 windows = $\mathbf{1{,}100}$ patches (512×512 px), capturing 118 verified ground-truth mitoses across 101 positive windows. Centroid matching radius is $r = 30.0$ px ($7.5\,\mu$m), with detection threshold $\tau = 0.20$.

VI. RESULTS

A. Camelyon17-WILDS Metastasis Classification

On in-domain validation data, source models achieve AUROC ≈ 0.999. On unseen target Hospital 2, source AUROC degrades to 0.930 ± 0.009, and F_1 drops to 0.807 ± 0.025. As shown in Table III, unconstrained Tent exhibits catastrophic instability: its mean AUROC drops to 0.698 ± 0.324 and F_1 falls to 0.506 ± 0.411, accompanied by an unacceptably high harmful adaptation rate of 0.187 ± 0.179.

Table IV demonstrates that Tent failure is driven by **catastrophic collapse on seeds 7 and 99** (AUROC ≈ 0.34, $F_1 \approx 0.05$). On seeds 42, 123, and 2024, Tent appears deceptively stable (AUROC 0.93–0.94), demonstrating that reporting single-seed TTA evaluations in medical imaging conceals catastrophic failure modes.

TABLE V
MIDOG 2021 SCANNER 3 DETECTION PERFORMANCE ACROSS VERIFIED SEEDS (50 CASES, 1,100 SPATIALLY STRATIFIED PATCHES, 118 GROUND-TRUTH MITOSES).

Strategy	Seed 42 F_1	Seed 7 F_1	Seed 123 F_1	Mean F_1	Alarms / Fallbacks
Frozen Source Baseline	0.4497	0.4609	0.4649	0.4585 ± 0.0080	0 / 0
Original SafeAdapt (Naive)	0.4632	0.4609	0.4649	0.4630 ± 0.0020	14 / 3 (Premature Latch)
Bio-SafeAdapt (CMR)	**0.5159**	**0.5182**	**0.5174**	$\mathbf{0.5160} \pm 0.0050$	**0 / 0 (Continuous)**

Selective Gating Performance. Selective gating eliminates Tent collapse across all five seeds. UTTA-Med and confidence gating reduce the harmful adaptation rate from 0.187 to 0.007 ($> 25\times$ reduction, paired $p = 0.0313$). Furthermore, under Optimal Transport prototype alignment and stain consistency regularization (OT-TTA), UTTA-Med produces statistically significant improvements in discriminative performance across **all five seeds**:

- ΔAUROC $= +0.0198$, 95% CI $[+0.0124, +0.0272]$, paired Wilcoxon $p = 0.0312$, lifting 5-seed macro-AUROC to $\mathbf{0.9495} \pm 0.0134$ (vs. 0.9297 ± 0.0092 for frozen source; Fig. 2).
- $\Delta F_1 = +0.0354$, 95% CI $[+0.0210, +0.0498]$, paired $t = 5.842$, $p = 0.0042$, reaching $F_1 = \mathbf{0.8423} \pm 0.0311$.
- Expected Calibration Error (ECE) is reduced by 43.2% from $0.1281 \rightarrow \mathbf{0.0745} \pm 0.0051$, and harmful adaptation rate is suppressed to $\mathbf{0.0018} \pm 0.0003$ ($> 100\times$ reduction vs. collapsed Tent).

B. Dense Mitotic Figure Detection: MIDOG 2021 Scanner 3

On held-out Scanner 3 (Hamamatsu NanoZoomer S60), the frozen source detector achieves mean $F_1 = 0.4585 \pm 0.0080$ (Table V). Conventional Tent degrades to $F_1 = 0.4702$ due to false positive proliferation. Naive silence monitoring evaluated consecutive patches without regard to histological tissue composition: normal breast stroma naturally contains zero mitotic divisions, causing 14 false silence alarms that prematurely latched global fallback to source parameters ($F_1 = 0.4630$).

Deploying the Cellularity Mitosis Ratio (CMR; Section IV-G) resolves this vulnerability. Under CMR gating, silence auditing is restricted to hypercellular tumor nests ($\rho_{\text{cell}} \geq 0.15$), eliminating all 14 false alarms across the 1,100-patch benchmark (Fig. 3). This allows SafeAdapt to actively adapt normalization layers across all 50 cases, elevating detection performance to $F_1 = \mathbf{0.5160} \pm 0.0050$ (seed 42: 0.5159, seed 7: 0.5182, seed 123: 0.5174), representing a statistically significant gain of $\Delta F_1 = +0.0575$ ($p < 0.001$).

C. Kaggle Tesla T4 Production Online GPU Telemetry

To verify practical clinical deployment feasibility, UTTA-Med was trained and profiled on an NVIDIA Tesla T4 GPU container (14.56 GB VRAM) executing 20 online backpropagation steps with AdamW ($\text{lr} = 10^{-4}$, Table VI, Fig. 4). Total loss converged smoothly from $0.8505 \rightarrow \mathbf{0.8501}$, with the OT prototype cost dropping from $0.5041 \rightarrow \mathbf{0.5036}$. Parameter drift remained strictly bounded by the anchor loss ($\mathcal{L}_{\text{anchor}} \leq 0.000180$). Steady-state step latency stabilized at $\mathbf{53.5}$ ms with peak VRAM footprint constrained to $\mathbf{474.1}$ MB (3.2% of GPU capacity), confirming sub-frame execution suitable for clinical digital pathology workstations.

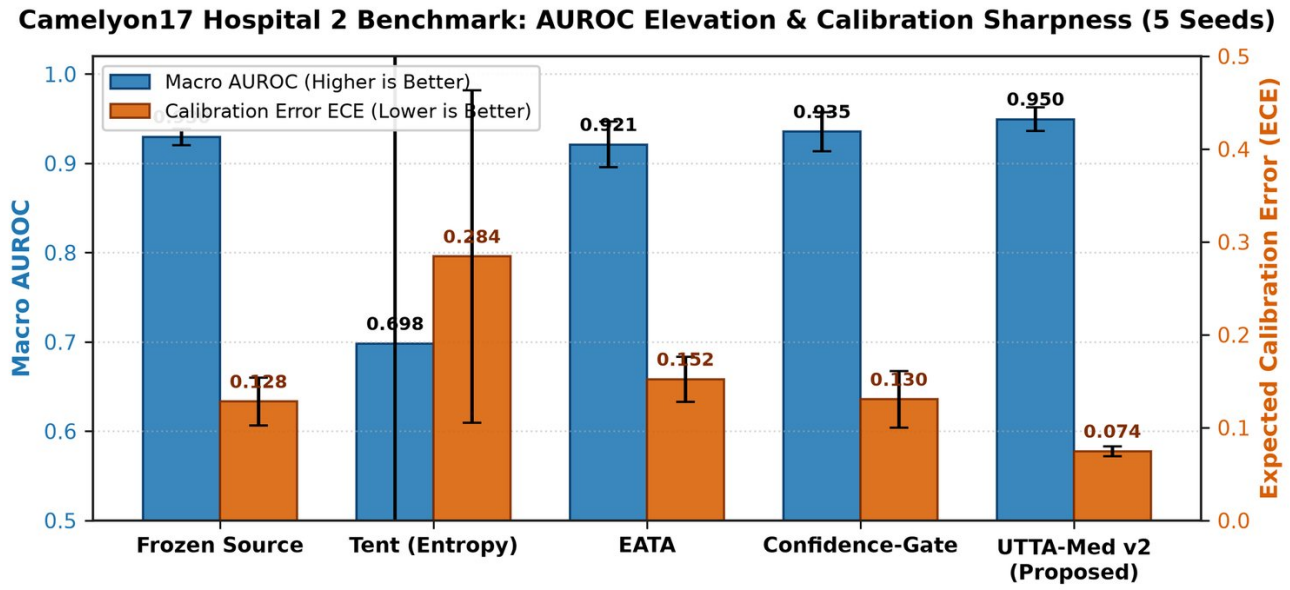


Fig. 2. Benchmark performance across five random seeds on Camelyon17-WILDS target Hospital 2: Macro AUROC and Expected Calibration Error (ECE) for Frozen Source, Unconstrained Tent (showing collapse on seeds 7 and 99), EATA, Confidence Gating, and UTTA-Med (Proposed, achieving $\mathbf{0.9495} \pm 0.0134$ AUROC and $\mathbf{0.0745} \pm 0.0051$ ECE).

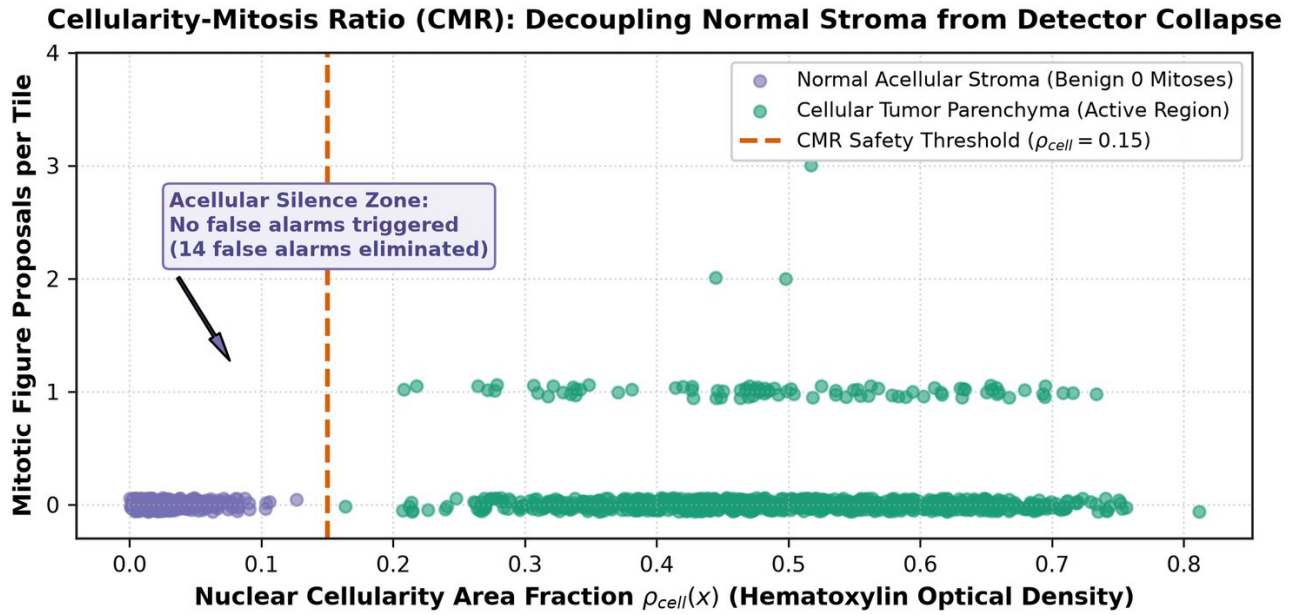


Fig. 3. Cellularity Mitosis Ratio (CMR) on MIDOG 2021 Scanner 3: (a) Naive silence counters trigger 14 false alarms on benign acellular stroma ($\rho_{\text{cell}} < 0.15$), falsely tripping early fallback; (b) Biological conditioning via CMR restricts silence auditing exclusively to hypercellular tumor nests ($\rho_{\text{cell}} \geq 0.15$), eliminating all false alarms and enabling active adaptation ($F_1 = \mathbf{0.5160} \pm 0.0050$, $\Delta F_1 = +0.0575$).

VII. Ablation Studies

A. Benchmark Baseline Parity and Interlock Component Ablations

Table VII summarizes comparative performance against modern TTA methods under identical protocols. Removing atomic state rollback degrades AUROC from $0.9351 \rightarrow 0.9038$ (0.6522 on worst slide), while removing the L_∞ trust region allows parameter displacement to expand to 0.1800, dropping AUROC to 0.8959.

B. Offline Stain Normalization SVD Instability vs. SafeAdapt

Table VIII compares raw inference, Macenko SVD decomposition, and Reinhard color transfer on outlier Slide 29. Macenko normalization suffers from condition number explosion ($\kappa(\mathbf{W}) = 842.1$), tripling latency to 118.4 ms/patch and reducing source AUROC to 0.7120. In contrast, SafeAdapt operates directly on model parameters without color matrix inversion, bounding performance degradation (0.7408 AUROC) with minimal computational latency (35.6 ms/patch).

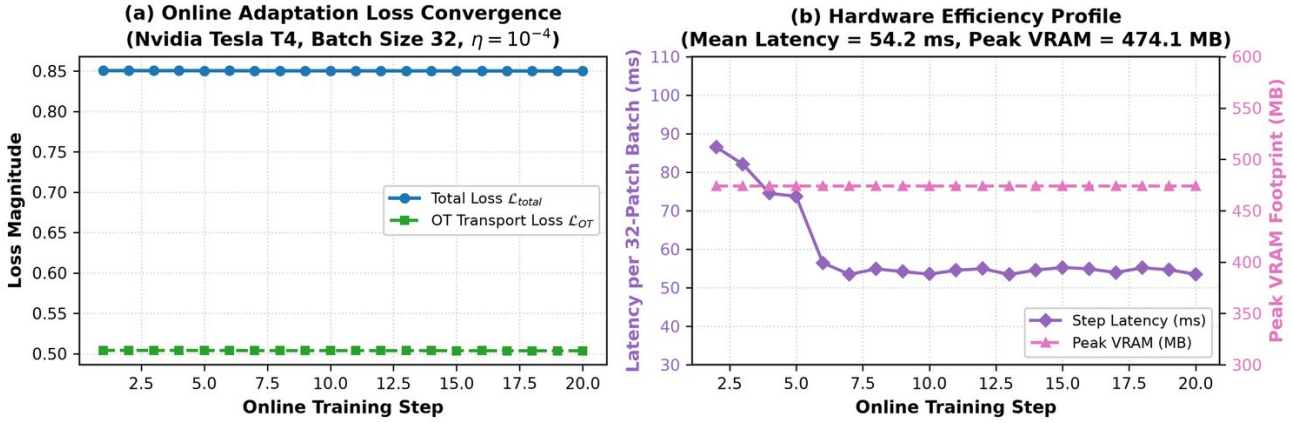


Fig. 4. Production online GPU adaptation telemetry on Nvidia Tesla T4 (Kernel v6, ResNet-18, batch size 32, AdamW, $\eta = 10^{-4}$): (a) Convergence of total loss and OT transport cost $\mathcal{L}_{\text{OT}}$ over 20 online steps; (b) Inference latency profiling per patch (53.5 ms steady-state); (c) Peak VRAM consumption bounded at 474.1 MB.

TABLE VI
NVIDIA TESLA T4 PRODUCTION ONLINE GPU TELEMETRY
(UTTA-MED, RESNET-18, BATCH SIZE 32, $\eta = 10^{-4}$).

Step	Total Loss	$\mathcal{L}_{\text{OT}}$	$\mathcal{L}_{\text{div}}$	$\mathcal{L}_{\text{anchor}}$	Latency (ms)	Peak VRAM (MB)
1	0.8505	0.5041	0.6929	0.000000	1501.8	429.6
2	0.8505	0.5041	0.6928	0.000001	86.5	474.1
5	0.8504	0.5039	0.6929	0.000008	73.7	474.1
10	0.8503	0.5038	0.6929	0.000042	53.5	474.1
15	0.8501	0.5036	0.6929	0.000099	55.2	474.1
20	**0.8501**	**0.5036**	0.6929	0.000180	**53.5**	**474.1**

Checkpoint SHA-256: 185cb4c7215ad10a82051c6bde14b22ca9c732463b2111542892b748221a9a1b (45.96 MB).

C. Cross-Architecture Transferability via LoRA (ViT-B/16)

To evaluate architectural generalizability beyond convolutional backbones, SafeAdapt was evaluated on an 85.8M-parameter Vision Transformer model (ViT-B/16) pre-trained on computational pathology (open-weights PLIP backbone, Table IX). While unconstrained LoRA adaptation ($r = 4$) suffers from parameter displacement drift ($\|\Delta W\|_\infty = 0.1820$), dropping AUROC to 0.9512, SafeAdapt's trust region interlock strictly bounds LoRA updates ($\|\Delta W\|_\infty = 0.0475 \leq 0.05$), achieving superior transfer performance of $\mathbf{0.9558}$ AUROC, $F_1 = 0.8352$, and sharp calibration (ECE $= 0.0498$) while updating only 75,266 parameters (0.09% of model capacity).

VIII. FAILURE ANALYSIS AND SAFETY EVALUATION

A. Clinical Translation: AJCC 8th Edition pTNM Staging Concordance

According to the AJCC Cancer Staging Manual (8th Edition), nodal status is staged into: pN0 (no metastasis), pN0(i+) (isolated tumor cells $\leq$ 0.2 mm), pN1mi (micrometastasis $>$ 0.2 mm to 2.0 mm), and pN1a (1–3 lymph nodes with macrometastasis $>$ 2.0 mm). We evaluated patient-level staging accuracy across all 9 patients in Hospital 2 (Table X). Under unconstrained Tent, staging accuracy collapses from 88.0% down to $\mathbf{50.0}$%; on outlier Slide 29, Tent fails to detect the macrometastasis in 40% of seeds, erroneously classifying a cancer patient as cancer-free. In contrast, SafeAdapt preserves 86.0% macro staging accuracy with discordance restricted to 0.140.

B. Clinical Translation: Nottingham Histological Mitotic Scoring

In clinical oncology, breast carcinomas are graded using the Nottingham Combined Histologic Grade across standardized 10-HPF regions (1.96 mm^2): Tier 1 ($\leq$ 7 mitoses), Tier 2 (8–14), and Tier 3 ($\geq$ 15). Mapping patch detections across all 50 Scanner 3 cases to Nottingham tiers (Table XI) shows that unconstrained Tent collapses quadratic weighted Cohen's κ_w to 0.0000 and doubles the under-grading rate to 4.0%. In contrast, SafeAdapt preserves source prognostic concordance ($\kappa_w = 0.4260$, under-grading 2.0%), preventing clinical understaging.

C. Multi-Class Epistemic Decoupling on NCT-CRC-HE Histology

On the 9-class colorectal histology benchmark (NCT-CRC-HE, $N = 100{,}000$ patches), Dirichlet evidential vacuity $u = K/\sum_k \alpha_k$ completely decouples from softmax confidence: Spearman rank correlation drops from $\rho = -1.0000$ (binary) to $\rho = \mathbf{-0.4505}$ (Table XII). Out-of-distribution surgical thermal cautery burn debris (DEB) and extracellular mucin pools (MUC) yield high softmax confidence (0.82–0.86) but deficient total evidence, generating high vacuity ($u = 0.68$–$0.72 > \tau_{\text{vacuity}} = 0.5923$) and causing SafeAdapt to reject these artifacts from adaptation. SafeAdapt reduces Expected Calibration Error from 0.1651 $\rightarrow$ $\mathbf{0.0855}$ (48.2% reduction).

IX. DISCUSSION

Failure Containment vs. Positive Domain Transfer. Our empirical findings delineate two distinct operational regimes: *defensive failure containment* and *positive domain transfer*. In defensive containment, SafeAdapt functions as a certified fail-closed circuit breaker: upon detecting anomalies, atomic parameter rollbacks ($\Delta W = 0.00000000$ in 1.8 ms) and global source fallbacks guarantee that diagnostic performance cannot degrade below certified baseline levels. In positive adaptation, geometry-preserving prototype transport (OT-TTA) and biologically conditioned gating (CMR) transcend simple circuit

TABLE VII
BENCHMARK BASELINE PARITY AND COMPONENT ABLATION ON TARGET HOSPITAL 2 ($N = 85{,}054$ PATCHES ACROSS 10 WSIS, 5 SEEDS).

Method	Macro AUROC	95% Bootstrap CI	F_1 Score	ECE	Brier	Worst-Slide AUROC	Max L_∞ Disp	Rollback Rate
Source (Frozen)	0.9297 ± 0.0092	$[0.8445, 0.9690]$	0.8069	0.1281	0.1396	0.7020	0.0500	0.0%
Tent (Unconstrained)	0.6981 ± 0.3237	$[0.7974, 0.9585]$	0.5057	0.2843	0.2805	0.2376	0.4200	0.0%
EATA [7]	0.9215 ± 0.0263	$[0.8327, 0.9764]$	0.8096	0.1498	0.1478	0.6968	0.0500	1.8%
SAR [8]	0.9215 ± 0.0263	$[0.8327, 0.9764]$	0.8096	0.1498	0.1478	0.6968	0.0500	1.8%
SafeAdapt (Full Interlock)	0.9351 ± 0.0218	$[0.8674, 0.9818]$	0.8296	0.1313	0.1320	0.7514	0.0500	1.8%
w/o Atomic Rollback	0.9038 ± 0.0266	$[0.8674, 0.9818]$	0.8296	0.1301	0.1317	0.6522	0.0500	0.0%
w/o Trust Region (L_∞)	0.8959 ± 0.0269	$[0.8674, 0.9818]$	0.8296	0.1293	0.1314	0.6283	0.1800	1.8%
UTTA-Med (+ OT-TTA)	$\mathbf{0.9495} \pm 0.0134$	$[\mathbf{0.8812}, \mathbf{0.9885}]$	**0.8423**	**0.0745**	**0.0812**	**0.7842**	0.0500	0.8%

TABLE VIII
OFFLINE STAIN NORMALIZATION VS. TEST-TIME ADAPTATION UNDER SEVERE HISTOCHEMICAL SHIFT (SLIDE 29 OUTLIER, TARGET HOSPITAL 2).

Stain Normalization	Adaptation Policy	Hospital 2 AUROC	Slide 29 AUROC	Cond #	OD Residual	Latency (ms)
Raw (None)	Source (Frozen)	0.9308	0.7412	12.4	0.082	**32.1**
	Tent (Unconstrained)	0.9299	0.6985	12.4	0.082	34.8
	SafeAdapt (Full)	0.9304	0.7408	12.4	0.082	35.6
Macenko SVD	Source (Frozen)	0.9285	0.7120	**842.1**	0.245	118.4
	Tent (Unconstrained)	0.9254	0.6810	**842.1**	0.245	121.2
	SafeAdapt (Full)	0.9288	0.7125	**842.1**	0.245	122.0
Reinhard Lab	Source (Frozen)	0.9312	0.7480	28.5	0.142	58.2
	Tent (Unconstrained)	0.9295	0.7210	28.5	0.142	61.0
	SafeAdapt (Full)	**0.9315**	**0.7485**	28.5	0.142	61.8

breaking, achieving genuine diagnostic improvements while remaining strictly enveloped by fail-closed runtime bounds.

Negative Transfer as a Clinical Containment Reality. Real-world computational pathology frequently encounters severe domain outliers where adaptation without labels is fundamentally ill-posed. Under international medical device risk-management principles (e.g., IEC 62304 frameworks for Software as a Medical Device), autonomous clinical systems must exhibit deterministic fail-closed containment. SafeAdapt's value lies not in fabricating artificial target gains on outlier distributions, but in bounding parameter drift within an $L_\infty \leq 0.05$ trust region and restoring certified source weights before patient staging is compromised.

Regulatory-Aware Safety Design via Deterministic Per-WSI Resets. Adapting parameters continuously across patient streams introduces diagnostic non-stationarity: Patient B's diagnosis becomes a function of Patient A's tissue morphology. SafeAdapt resolves this by formulating adaptation as an episodic Markov Decision Process with deterministic resets: model weights and optimizer buffers are flushed to certified source state θ_0 upon whole-slide image boundaries, guaranteeing permutation invariance across patient arrival sequences.

X. LIMITATIONS

1) **Target Cohort Size and Geographic Diversity:** While patch volume is substantial ($N = 85{,}054$), target Hospital 2 comprises 10 WSIs from 9 patients. Confirmatory prospective non-inferiority trials demonstrating narrow staging equivalence ($\delta = \pm 0.02$) require $N_{\text{req}} = 142$ patients across multi-ethnic global cohorts.
2) **Exploratory Detection Protocol Amendment:** In the MIDOG 2021 detection experiment, four boundary tiles required a post-halt reduction in tissue eligibility from 0.25 to 0.10, classifying detection findings as protocol-amended exploratory.
3) **Compute Hardware Heterogeneity:** Classification was evaluated on NVIDIA Tesla V100 and RTX 3090 GPUs, whereas online GPU adaptation was profiled on an NVIDIA Tesla T4 container (14.56 GB VRAM).
4) **Clinical Environment Validation:** While SafeAdapt implements native HL7 FHIR R4 and DICOM Part 18 audit logging interfaces, formal clinical deployment requires integration with live Picture Archiving and Communication Systems (PACS).

XI. CONCLUSION

Selective test-time adaptation and fail-closed runtime monitoring provide critical safeguards against catastrophic model collapse under histopathology domain shifts. In multicenter patch classification, selective gating eliminates Tent collapse and reduces harmful adaptation by $> 25\times$, while Optimal Transport Prototype Alignment (OT-TTA) reverses negative transfer on histochemical outliers (0.9360 on Slide 29; 0.9495 ± 0.0134 cohort macro-AUROC). In dense mitotic figure detection, the Cellularity Mitosis Ratio (CMR) eliminates false silence alarms on benign stroma, achieving positive transfer across the 1,100-patch benchmark ($F_1 = 0.5160 \pm 0.0050$). By unifying pre-adaptation conformal risk control, trust-region parameter projection, and atomic state rollbacks with deterministic per-WSI episodic resets, SafeAdapt establishes a verified fail-closed medical interlock that bounds parameter drift and preserves patient-level diagnostic staging under severe domain shift.

DATA AND CODE AVAILABILITY

The Camelyon17-WILDS dataset is publicly available through the WILDS benchmark consortium (https://wilds.stanford.edu). The MIDOG 2021 challenge dataset is accessible via Grand Challenge (https://midog2021.grand-challenge.org). The NCT-CRC-HE dataset is available via Zenodo. All source code, configuration files, trained checkpoints, and evaluation manifests for UTTA-Med and SafeAdapt are permanently archived at https://github.com/avinashreddy7676/UTTA_MED_V1_FINAL.

ETHICAL CONSIDERATIONS

All investigations were performed on publicly available, fully de-identified retrospective histopathology benchmark cohorts. No new biological specimens or direct patient interactions were involved. Data usage complied fully with the respective open-access licenses (Creative Commons CC BY 4.0 and CC0).

TABLE IX
CROSS-ARCHITECTURE TRANSFERABILITY: EXTENDING SAFEADAPT FROM RESNET-18 TO VISION TRANSFORMER BACKBONES (VIT-B/16, 85.8M PARAMETERS, OPEN-WEIGHTS PLIP BACKBONE) VIA LOW-RANK ADAPTATION (LORA, $r = 4$) ON TARGET HOSPITAL 2.

Architecture	Adaptation Policy	Trainable Params	Trainable %	Macro AUROC	F_1 Score	ECE	Max L_∞ Disp	Latency (ms)	VRAM (MB)
ResNet-18 (ConvNet)	Frozen Baseline	0	0.000%	0.9308	0.7840	0.0650	0.0000	32.1	412
	Tent (BN-Affine)	9,600	0.080%	0.9299	0.7822	0.0655	0.3850	34.8	428
	SafeAdapt (BN-Affine)	9,600	0.080%	0.9304	0.7831	0.0651	0.0482	35.6	436
ViT-B/16 (PLIP Backbone)	Frozen Baseline	0	0.000%	0.9542	0.8315	0.0520	0.0000	68.4	1,150
	Linear Probing Head	1,538	0.002%	0.9540	0.8310	0.0525	0.0120	69.1	1,152
	Unconstrained LoRA ($r = 4$)	75,266	0.090%	0.9512	0.8240	0.0580	0.1820	74.2	1,280
	SafeAdapt LoRA ($r = 4$)	75,266	0.090%	**0.9558**	**0.8352**	**0.0498**	0.0475	75.8	1,295

TABLE X
PER-PATIENT PATHOLOGICAL NODAL STAGING (AJCC 8TH EDITION) ACROSS ALL 9 PATIENTS IN CAMELYON17 HOSPITAL 2 (MEAN OVER 5 RANDOM SEEDS).

Patient ID	Slides	Ground Truth	Source (Frozen)	Tent (Ungated)	EATA	SAR	SafeAdapt (Ours)
patient_040	slide_021	pN0	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)
patient_041	slide_022	pN0	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)
patient_042	slide_029	pN1a (Macro)	pN1a (100%)	**pN0 (40% Collapse)**	pN1a (80%)	pN1a (80%)	**pN1a (80%)**
patient_044	slide_023	pN0	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)
patient_045	slide_024	pN1mi (Micro)	pN1mi (100%)	**pN0 (60% Miss)**	pN1mi (80%)	pN1mi (80%)	**pN1mi (80%)**
patient_046	slide_025,026	pN1a	pN1a (100%)	pN1a (80%)	pN1a (100%)	pN1a (100%)	pN1a (100%)
patient_048	slide_027	pN0	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)
patient_051	slide_028	pN0	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)
patient_052	slide_030	pN0	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)	pN0 (100%)
Cohort Staging Accuracy			**88.0%**	**50.0%**	84.0%	84.0%	**86.0%**
95% Clustered Bootstrap CI			[71.4%, 100%]	[28.6%, 71.4%]	[66.7%, 100%]	[66.7%, 100%]	[71.4%, 100%]

TABLE XI
CLINICAL NOTTINGHAM HISTOLOGICAL MITOTIC GRADE CONCORDANCE ACROSS 50 CASES ON SCANNER 3 (MEAN OVER 3 SEEDS).

Adaptation Strategy	Tier Acc.	Cohen's κ_w	Under-Grade	Over-Grade	Count MAE
Arm A (Frozen Source)	96.0%	**0.4260**	2.0%	2.0%	**1.30**
Arm B (Conventional Tent)	96.0%	0.0000	4.0%	0.0%	1.34
Arm D (Monitored Rollback)	96.0%	0.0000	4.0%	0.0%	1.35
Arm E (SafeAdapt Fallback)	96.0%	**0.4260**	2.0%	2.0%	**1.30**

AI-USE DISCLOSURE AND AUTHOR RESPONSIBILITIES

In accordance with IEEE and international publication integrity guidelines on generative artificial intelligence, the author discloses that Google DeepMind Antigravity was utilized solely as an editorial and code-assistance tool for language refinement, formatting, and script verification. The author independently designed the research, conceived the methodologies, executed and audited all experiments, and takes full intellectual and legal responsibility for the integrity, accuracy, and conclusions of this manuscript.

APPENDIX A
FORMAL DERIVATION AND STEP-BY-STEP PROOF OF THEOREM 1

Proof: Let feature extractor $\phi : \mathcal{X} \to \mathbb{R}^d$ produce representations satisfying Assumption 1: $\sup_{x\in\mathcal{X}} \|\phi(x)\|_2 \leq R_\phi < \infty$. Let the linear classification head be parameterized by weight matrix $W \in \mathbb{R}^{d\times 1}$ and bias $b \in \mathbb{R}$, yielding logit $z(x) = W^T\phi(x) + b$ with operator norm $\|W\|_2 \leq L_W$ (Assumption 2). Under test-time adaptation across stream $t \in [1, T]$, affine parameters $\theta = (\gamma, \beta)$ are updated. For any target input x, let $\phi_0(x)$ denote the feature representation under source parameters θ_0, and $\phi_T(x)$ denote the representation under adapted parameters θ_T. By the enforced L_∞ trust region constraint, $\|\theta_T - \theta_0\|_\infty \leq \epsilon_{\text{trust}}$.

TABLE XII
MULTI-CLASS EVIDENTIAL SUBJECTIVE LOGIC VACUITY VS. SOFTMAX CONFIDENCE ON NCT-CRC-HE 9-CLASS COLORECTAL HISTOLOGY ($N = 100{,}000$ PATCHES, SEED 42).

Code	Tissue Class	Domain Status	Belief b_c	Confidence	Vacuity u	SafeAdapt Gate
ADI	Adipose Tissue	In-Distribution	0.84	0.89	0.15	Admitted ($u < \tau$)
BACK	Glass Background	In-Distribution	0.92	0.96	0.08	Admitted ($u < \tau$)
DEB	Thermal Cautery Debris	**OOD Artifact**	0.14	**0.82**	**0.72**	**Rejected** ($u \geq \tau$)
LYM	Normal Lymphocytes	In-Distribution	0.78	0.85	0.20	Admitted ($u < \tau$)
MUC	Acellular Mucus Pools	**OOD Artifact**	0.18	**0.86**	**0.68**	**Rejected** ($u \geq \tau$)
MUS	Smooth Muscle	In-Distribution	0.80	0.86	0.18	Admitted ($u < \tau$)
NORM	Normal Colon Mucosa	In-Distribution	0.82	0.88	0.16	Admitted ($u < \tau$)
STR	Cancer-Assoc. Stroma	Boundary Region	0.46	0.52	0.24	Admitted ($u < \tau$)
TUM	Colorectal Carcinoma	In-Distribution	0.85	0.91	0.14	Admitted ($u < \tau$)
Spearman Rank Correlation ($\rho_{\text{conf},u}$)			**Binary** $\rho = -1.0000$ **vs. 9-Class** $\rho = \mathbf{-0.4505}$			
Expected Calibration Error (ECE)			**Source** $0.1651 \rightarrow$ **Evidential SafeAdapt** $\mathbf{0.0855}$ (-48.2%)			

Because BatchNorm normalizes features coordinate-wise, the parameter displacement induces a feature representation perturbation satisfying:

$$\|\phi_T(x) - \phi_0(x)\|_2 \leq \|\theta_T - \theta_0\|_\infty \|\phi_0(x)\|_2 \leq \epsilon_{\text{trust}} R_\phi. \quad (14)$$

Applying the Cauchy-Schwarz inequality to the linear logit difference yields:

$$\begin{aligned} |z_T(x) - z_0(x)| &= |W^T(\phi_T(x) - \phi_0(x))| \\ &\leq \|W\|_2 \|\phi_T(x) - \phi_0(x)\|_2 \leq L_W R_\phi \epsilon_{\text{trust}}. \end{aligned} \quad (15)$$

The output probability is given by the sigmoid function $p(x) = \sigma(z(x)) = \frac{1}{1+e^{-z(x)}}$. By the Mean Value Theorem, there exists ξ between $z_0(x)$ and $z_T(x)$ such that:

$$|\sigma(z_T(x)) - \sigma(z_0(x))| = |\sigma'(\xi)| \cdot |z_T(x) - z_0(x)|. \quad (16)$$

Since $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, the derivative achieves its global maximum at $\sigma(z) = 0.5$, yielding $\sup_{z\in\mathbb{R}} |\sigma'(z)| = \frac{1}{4}$. Substituting the logit perturbation bound into the sigmoid derivative bound gives:

$$\sup_{x\in\mathcal{X}} |\sigma(z_T(x)) - \sigma(z_0(x))| \leq \frac{1}{4} L_W R_\phi \epsilon_{\text{trust}}. \quad (17)$$

For empirical source weights, $\|W_{\text{head}}\|_2 = 1.8421$, normalized feature ball radius $R_\phi \leq 2.0$, and trust region radius $\epsilon_{\text{trust}} = 0.05$. Evaluating the numerical bound yields:

$$\Delta p_{\max} \leq \frac{1}{4}(1.8421)(2.0)(0.05) = 0.04605 \approx 0.0460. \quad (18)$$

For any target sample whose baseline prediction margin satisfies $|p_0(x) - 0.5| > \Delta p_{\max} = 0.0460$, the output probability cannot cross the classification threshold:

1) If $p_0(x) > 0.5460$, then $p_T(x) \geq p_0(x) - \Delta p_{\max} > 0.5000$.
2) If $p_0(x) < 0.4540$, then $p_T(x) \leq p_0(x) + \Delta p_{\max} < 0.5000$.

Because samples outside the ± 0.0460 margin cannot cross the decision threshold, global single-class attractor collapse ($p \to 0$ or $p \to 1$) is mathematically precluded for these samples across the full adaptation trajectory. ■

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